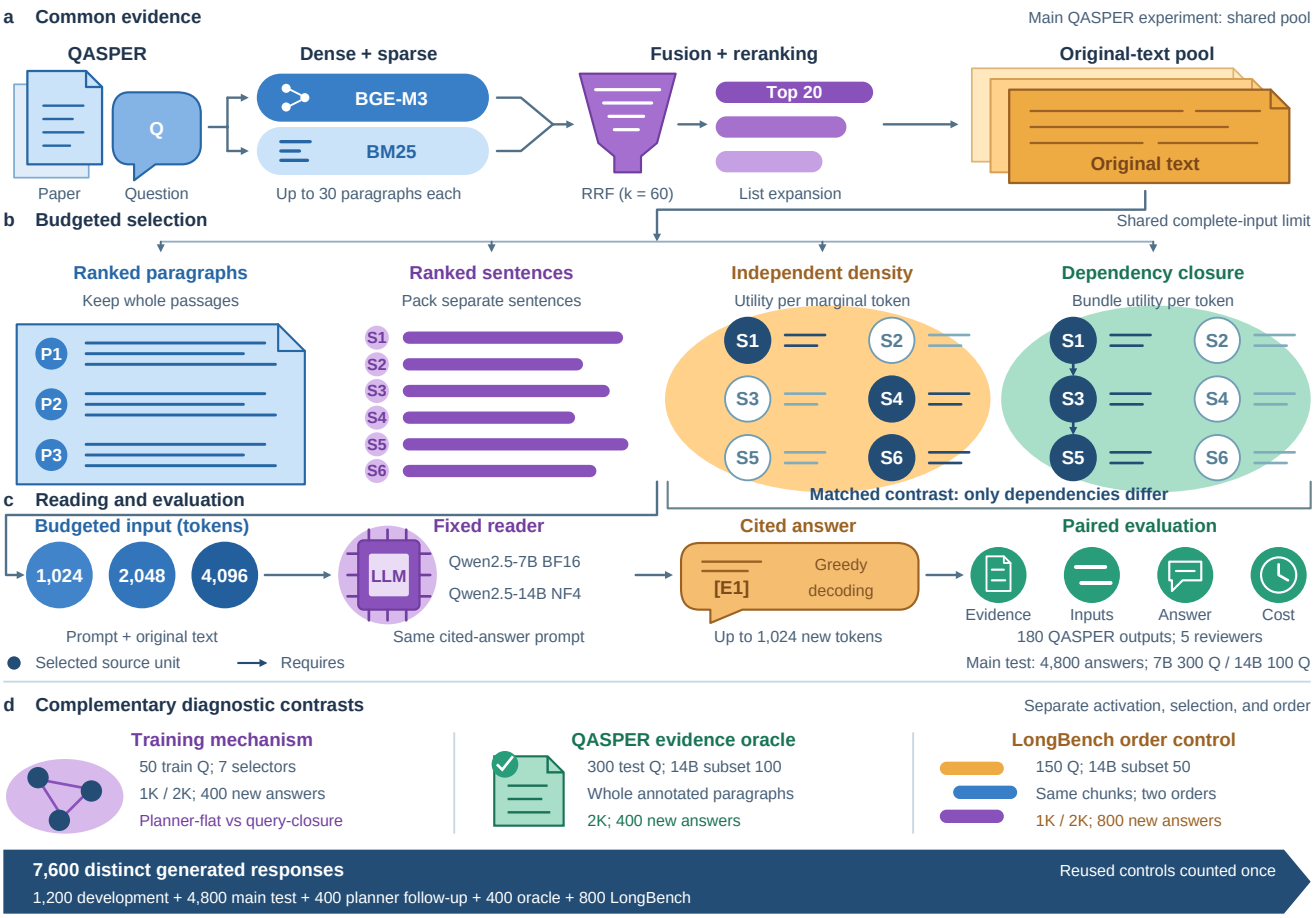
a Common evidence
Main QASPER experiment: shared pool
QASPER
Dense + sparse
Fusion + reranking
Original-text pool
Q
BGE-M3
BM25
Top 20
Original text
Paper
Question
Up to 30 paragraphs each
RRF (k = 60)
List expansion
b Budgeted selection
Shared complete-input limit
Ranked paragraphs
Keep whole passages
Ranked sentences
Pack separate sentences
Independent density
Utility per marginal token
Dependency closure
Bundle utility per token
P1
P2
P3
S1
S2
S3
S4
S5
S6
Matched contrast: only dependencies differ
c Reading and evaluation
Budgeted input (tokens)
1,024
2,048
4,096
Prompt + original text
LLM
Fixed reader
Qwen2.5-7B BF16
Qwen2.5-14B NF4
Same cited-answer prompt
Cited answer
[E1]
Greedy decoding
Up to 1,024 new tokens
Paired evaluation
Evidence
Inputs
Answer
Cost
180 QASPER outputs; 5 reviewers
Selected source unit
Requires
Main test: 4,800 answers; 7B 300 Q / 14B 100 Q
d Complementary diagnostic contrasts
Separate activation, selection, and order
Training mechanism
50 train Q; 7 selectors
1K / 2K; 400 new answers
Planner-flat vs query-closure
QASPER evidence oracle
300 test Q; 14B subset 100
Whole annotated paragraphs
2K; 400 new answers
LongBench order control
150 Q; 14B subset 50
Same chunks; two orders
1K / 2K; 800 new answers
7,600 distinct generated responses
1,200 development + 4,800 main test + 400 planner follow-up + 400 oracle + 800 LongBench
Reused controls counted once